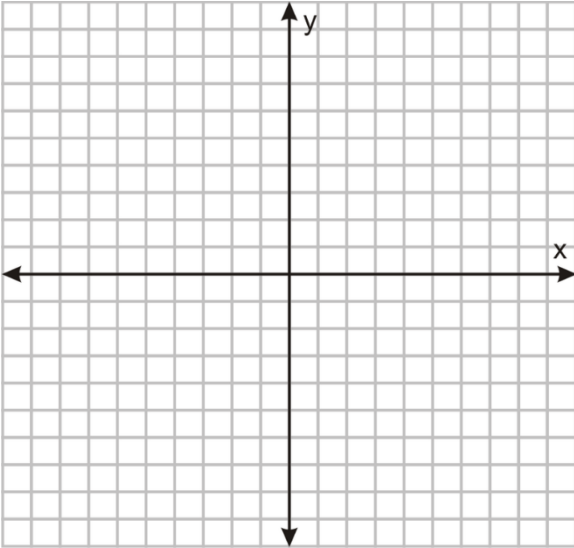


Homework: Rational Piecewise Functions

I. Analyzing the parent function:

1. Graph: $f(x) = \frac{1}{x}$



Domain:

Range:

Increasing:

Decreasing:

End behavior:

II. Rational Piecewise Functions:

$$f(x) = \begin{cases} -\frac{1}{2}x - 1 & , \quad x \leq 0 \\ \frac{1}{x+1} - 2 & , \quad x > 0 \end{cases}$$

Domain:

Range:

Increasing:

Decreasing:

End behavior:

Continuous?

